

1. What is the output of the following code?

```
#include <iostream>
using namespace std;

int gcd(int x, int y)
{
    if (x % y == 0)
        return y; // What case is this?
    else
        return gcd(y, x % y); // What case is this?
}

int main()
{
    cout << gcd(20, 8) << endl;
    return 0;
}
```

2. What is the output of the following program.

```
#include <iostream>
using namespace std;

int function(int num) {
    if (num <= 0)
        return 0;
    else
        return function(num - 1) + num;
}

int main() {
    cout << function(10) << endl;
}
```

3. What is the output of the following code?

```
#include <iostream>
using namespace std;

void function(int num) {
    if (num > 0)
    {
        for (int x = 0; x < num; x++)
            cout << '*';

        cout << endl;
        function(num - 1);
    }
}

int main() {
    function(10);
}
```

4. What is the output of the following code?

```
void function(string, int, int);

int main()
{
    string mystr = "Hello";
    cout << mystr << endl;
    function(mystr, 0, mystr.size());
    return 0;
}

void function(string str, int pos, int size)
{
    if (pos < size)
    {
        function(str, pos + 1, size);
        cout << str[pos];
    }
}
```

5. Convert the following function to one that uses recursion. How would the driver program be written for the recursive function?

```
void sign(int n)
{
    while (n > 0)
        cout << "No Parking\n";
    n--;
}
```

6. Answer the following questions about the following code.

```
int recFunc(int num)
{
    if (num == 0)
        return 0;
    else if (num < 0)
        return (-num);
    else
        return (num - recFunc(num - 5));
}
```

- Identify the base case and the recursive case.
- If `recFunc(20)` is a valid call, what is its value? If not, explain why.

7. Consider the following recursive function:

```
void funcEx8(int u, char v)
{
    if (u == 0)
        cout << u << " ";
    else
    {
        int x = static_cast<int>(v);
        if (v < 'A')
            v = static_cast<char>(x + 1);
        else if (v > 'Z')
            v = static_cast<char>(x - 1);
        cout << v << " ";
        funcEx8(u - 2, v);
    }
}
```

Answer the following questions:

- Identify the base and recursive case.
- If `funcEx8(6, '$')` is a valid call, what is the output? If not, explain why.

8. Consider the following recursive function:

```
void recFun(int x)
{
    if (x > 0)
    {
        cout << x % 10 << " ";
        recFun(x / 10);
    }
    else if (x != 0)
        cout << x << endl;
}
```

What is the output of the following statements?

- a) recFun(36);

9. Consider the following recursive function:

```
void recFun(int u)
{
    if (u == 0)
        cout << "Zero! ";
    else
    {
        cout << "Negative ";
        recFun(u - 1);
    }
}
```

What is the output, if any, of the following statements?

- a) recFun(8);

10. Consider the following recursive function:

```
void recEx11(int x){
    if (x > 0)
        if (x % 2 == 0)
        {
            recEx11((x - 3) / 2);
            cout << x << " ";
        }
        else
        {
            recEx11((x - 4) / 2);
            cout << x << " ";
        }
}
```

What is the output of the following statements?

- a) `recEx11(28);`
- b) `recEx11(-30);`

11. Consider the following function:

```
int recEx12(int a, int b)
{
    if (a < b)
        return a + b;
    else
        return recEx12(a - b, b + 1);
}
```

What is the output of the following statements?

- a) `cout << recEx12(128, 15) << endl;`

12. Consider the following function:

```
int rec(int x)
{
    if (x <= 0)
        return 0;
    else if (x == 1)
        return 2;
    else
        return (rec(x - 1) + rec(x - 2) + rec(x - 3));
}
```

What is the output of the following statements?

- a) `cout << rec(4) << endl;`

13. Consider the following recursive function:

```
void recFun(int x, int y)
{
    if (x > 0 && y > 0)
    {
        if (x >= y && y != 0)
        {
            cout << x % y << " ";
            recFun(x - y, y);
        }
        else if (y > x && x != 0)
        {
            cout << y % x << " ";
            recFun(y - x, x);
        }
    }
    else
        cout << x + y << endl;
}
```

What is the output of the following statements?

- a) recFun(75, 26);
- b) recFun(13, 86);

14. Consider the following recursive function:

```
int test(int x, int y)
{
    if (abs(x - y) <= 1)
        return x + y;
    else if (x > y)
        return test(x - 1, y + 1);
    else if (y > x)
        return test(x + 1, y - 1);
}
```

What is the output of the following statements? (4, 6)

- a. cout << test(8, 2) << endl;
- b. cout << test(5, 16) << endl;
- c. cout << test(-25, 2) << endl;
- d. cout << test(8, -6) << endl;
- e. cout << test(-20, -36) << endl

15. What is the output?

```
struct Node {
    int data;
    Node* next;
    Node(int val) : data(val), next(nullptr) {}
};

void printList(Node* head) {
    if (head == nullptr) return; // base case
    cout << head->data << " ";
    printList(head->next); // recursive case
}

int main() {
    Node* head = new Node(1);
    head->next = new Node(2);
    head->next->next = new Node(3);
    printList(head);
}
```

16. What is the output? Assume the operation is performed on the linked list created in the above question.

```
Node* reverseList(Node* head) {
    if (head == nullptr || head->next == nullptr)
        return head;
    Node* rest = reverseList(head->next);
    head->next->next = head;
    head->next = nullptr;
    return rest;
}
```